

Translated

From: e-maxx.ru

Finding repetitions

Given a string s of length n .

A **repetition** is two occurrences of a string in a row. In other words a repetition can be described by a pair of indices $i < j$ such that the substring $s[i \dots j]$ consists of two identical strings written after each other.

The challenge is to **find all repetitions** in a given string s . Or a simplified task: find **any** repetition or find the **longest** repetition.

The algorithm described here was published in 1982 by Main and Lorentz.

Example

Consider the repetitions in the following example string:

$s = \text{acababaaee}$

The string contains the following three repetitions:

- $s[2 \dots 5] = \text{abab}$
- $s[3 \dots 6] = \text{baba}$
- $s[7 \dots 8] = \text{ee}$

Another example:

$s = \text{abaabaa}$

Here there are only two repetitions

- $s[0 \dots 5] = \text{abaaba}$
- $s[2 \dots 3] = \text{aa}$